

MAHFUZUR RAHMAN

Graduate Aeronautical Engineer · Computational Solid Mechanics & Composite Structures

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RESEARCH INTERESTS

Computational solid mechanics, with a focus on the failure of composite and multi-material structures under coupled thermo-mechanical loading. Particular interests:

- Damage and fracture modelling in fibre-reinforced and metal-matrix composites.
- Thermo-mechanical coupling and stress evolution in anisotropic materials.
- Numerical methods — finite-difference and finite-element frameworks — for structural problems that resist closed-form solution.

EDUCATION

B.Sc. in Aeronautical Engineering (Aerospace) 2022 – 2026

Military Institute of Science and Technology (MIST), Dhaka, Bangladesh

CGPA 3.80 / 4.00 · Dean's List of Honour 2023 & 2024

Degree requirements complete; thesis defended. Provisional certificate pending.

Higher Secondary Certificate (HSC) — Science 2021

Chittagong College, Chattogram Board · GPA 5.00 / 5.00 · Board Scholarship (General)

Secondary School Certificate (SSC) — Science 2019

Chittagong Collegiate School, Chattogram Board · GPA 5.00 / 5.00 · Board Scholarship (Talentpool)

RESEARCH EXPERIENCE

Undergraduate Thesis 2025 – 2026

Thermomechanical Behaviour of Fiber-Reinforced Composite Materials

Supervisor: Prof. Dr. Shaikh Reaz Ahmed, Dept. of Mechanical Engineering, BUET

- Developed a finite-difference computational framework in MATLAB to investigate thermal stress distributions in orthotropic metal-matrix composite (MMC) panels under combined thermal and mechanical loading.
- Characterised how material anisotropy governs stress concentration and the location of failure-prone zones in panels representative of high-temperature aerospace structures.
- Validated the numerical model against ANSYS thermal-structural simulations and analysed agreement across loading regimes.

PUBLICATIONS / WORKS IN PROGRESS

"Thermal Stress Analysis of Anisotropic MMC Panels." Work in progress under the supervision of Prof. Dr. Shaikh Reaz Ahmed, Dept. of Mechanical Engineering, BUET.

TECHNICAL SKILLS

Simulation & CAD: MATLAB, ANSYS (Structural & Thermal), ABAQUS, SolidWorks, XFLR5, AVL

Numerical Methods: Finite Element Method, Finite Difference Method, numerical analysis, structural solvers

Mechanics: Composite mechanics, thermal stress analysis, structural analysis, fatigue & failure, aerodynamics

Documentation: LaTeX, technical report writing, Microsoft Office (Advanced)

Certifications: MATLAB Onramp (MathWorks), Simulink Onramp (MathWorks)

COMPETITIONS & ACHIEVEMENTS

AIAA Design/Build/Fly (DBF) 2026 — Team Lead

2025 – 2026

18th Worldwide — Proposal Report | 17th Worldwide — Design Report

- Led the MIST team through the full engineering lifecycle — aerodynamic sizing, structural design, composite fabrication, and flight testing.
- Authored the internationally peer-evaluated Proposal and Design Reports, both placing in the top 20 worldwide.

AIAA Design/Build/Fly (DBF) 2025 — Manufacturing Lead & Flyoff Member

2024 – 2025

Tucson, Arizona, USA | 1st in Asia — Total Flight Mission Score | 39th Worldwide Overall

- Led fabrication and structural assembly, ensuring dimensional accuracy and strict weight compliance throughout the build cycle.
- Served as Flyoff Member on competition day — pre-flight checks and real-time mission assessment.

INDUSTRIAL TRAINING

On-the-Job Training — US-Bangla Airlines Ltd. (CAMO & AMO)

May – Jun 2025

- Reviewed maintenance documentation and airworthiness directives for ATR 72-600 and Boeing 737-800 fleets, gaining grounded exposure to how structural integrity is managed in operational aviation.
- Introductory training in Non-Destructive Testing (NDT) and maintenance-compliance procedures.

LEADERSHIP & ACTIVITIES

MIST Aeronautics & Astronautics Club (MAAC): Vice President – Administration (Jun 2025–Jun 2026); Executive – Administration (Sep 2024–Jun 2025); Event Manager (May 2023–Sep 2024).

Department Representative — Aeronautical Engineering, MIST: Elected liaison between students, Head of Department, and the Directorate of Student Welfare (Jul 2025–Jun 2026).

MIST Drama & Film Society: Director of Event Management (2024–2025); HR & Treasurer (2023–2024).

TEACHING

Private Tutor — SSC & HSC Level

2021 – 2025

- Delivered one-on-one instruction across Mathematics, Higher Mathematics, Physics, Chemistry, Biology, and ICT over four years; developed structured communication of technical material.

HONOURS & AWARDS

- Dean's List of Honour — 2023 & 2024, MIST.
- MIST Merit Scholarship — Spring 2025.
- Board Scholarship (Talentpool) — SSC 2019; Board Scholarship (General) — HSC 2021.

LANGUAGES

Bangla — native. English — fluent (reading, writing, speaking).

REFERENCES

Prof. Dr. Shaikh Reaz Ahmed — Professor, Dept. of Mechanical Engineering, BUET; Thesis Supervisor. Contact available on request.

Saif Reza — Lecturer, Dept. of Aeronautical Engineering, MIST; saifreza@ae.mist.ac.bd

Md. Ridwan Hasan — Lecturer, Dept. of Aeronautical Engineering, MIST; ridwan@ae.mist.ac.bd